A STABILIZER PRESERVING ETALE MORPHISM IS AN OPEN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that a morphism whose diagonal is an isomorphism is universally injective. Invoked at the start of the counted proof (source line 10172). Gives universal injectivity from the hypothesis on the diagonal.
- The combination of stabilizer preservation with unramifiedness of the automorphism group. Invoked inside the counted proof (source lines 10173-10176). Makes the morphism representable by algebraic spaces.
- The statement that an etale universally injective morphism of algebraic spaces is an open immersion. Invoked near the end of the counted proof (source lines 10177-10179) as a cross-chapter pointer. Turns the two properties into an open immersion.

Source reference. Original lemma lemma-aut-iso-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10012–10012. Navigation pointer only: the counted text refers to the Stacks item lemma-aut-iso-unramified, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-stabilizer-preserving: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9980–9980. Navigation pointer only: the counted text refers to the Stacks item lemma-stabilizer-preserving, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-universally-injective: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2907–2907. Navigation

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

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Source reference. Original remark remark-identify-automorphism-groups: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4046–4046. Navigation pointer only: the counted text refers to the Stacks item remark-identify-automorphism-groups, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-universally-injective-open: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4656–4656. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-universally-injective-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Assume f is étale, f induces an isomorphism between automorphism groups at points (Remark remark-identify-automorphism-groups), and for every algebraically closed field k the functor*

$$f : \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}) \longrightarrow \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{Y})$$

is an equivalence. Then f is an isomorphism.

Proof. By Lemma lemma-universally-injective we see that f is universally injective. Combining Lemmas lemma-stabilizer-preserving and lemma-aut-iso-unramified we see that f is representable by algebraic spaces. Hence f is an open immersion by Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-universally-injective-open. To finish we remark that the condition in the lemma also guarantees that f is surjective. □